Q1 2021 (01 Sep Shift 2)

A carrier wave with amplitude of 250 V is amplitude modulated by a sinusoidal base band signal of amplitude 150 V. The ratio of minimum amplitude to maximum amplitude for the amplitude modulated wave is 50 : x, then value of x is ....

Q2 2021 (31 Aug Shift 2)

A bandwidth of 6MHz is available for A.M. transmission. If the maximum audio signal frequency used for modulating the carrier wave is not to exceed 6kHz. The number of stations that can be broadcasted within this band simultaneously without interfering with each other will be ....

Q3 2021 (31 Aug Shift 1)

If the sum of the heights of transmitting and receiving antennas in the line of sight of communication is fixed at 160 m, then the maximum range of LOS communication is .....km.

(Take radius of Earth $= 6400$ km)

Q4 2021 (27 Aug Shift 2)

An antenna is mounted on a 400 m tall building. What will be the wavelength of signal of signal that can be radiated effectively by the transmission tower upto a range of 44 km ?

(1) 37.8 m

(2) 605 m

(3) 75.6 m

(4) 302 m

Q5 2021 (27 Aug Shift 1)

A transmitting antenna has a height of 320 m and that of receiving antenna is 2000 m. The maximum distance between them for satisfactory communication in line of sight mode is 'd'. The value of 'd' is .... km.

Q6 2021 (26 Aug Shift 2)

A transmitting antenna at top of a tower has a height of 50 m and the height of receiving antenna is 80 m. What is range of communication for Line of Sight (LoS) mode ?

[use radius of earth = 6400 km]

(1) 45.5 km

(2) 80.2 km

(3) 144.1 km

(4) 57.28 km

Q7 2021 (26 Aug Shift 1)

An amplitude modulated wave is represented by $C_m(t) = 10(1 + 0.2\cos 12560t)\sin\left(111 \times 10^4 t\right)$ volts. The modulating frequency in kHz will be

Answer Key

Q1 (200) **Q2** (500) **Q3** (64) **Q4** (2)

Q5 (224) **Q6** (4) **Q7** (2)

Q1 (200)

$A_{\max}=A_C+A_m=250+150=400$

$A_{\min}=A_C-A_m=250-150=100$

$\frac{A_{\min}}{A_{\max}}=\frac{100}{400}=\frac{1}{4}=\frac{50}{200}$

$x=200$

Q2 (500)

Signal bandwidth $=2\mathrm{fm}$

$=12\mathrm{kHz}$

$\therefore \mathrm{N}=\frac{6\mathrm{MHZ}}{12\mathrm{kHZ}}=\frac{6\times10^6}{12\times10^3}=500$

Q3 (64)

$\mathrm{h_T}=\mathrm{h_R}=160\ldots\text{(i)}$

$\mathrm{d}=\sqrt{2\mathrm{Rh_T}}+\sqrt{2\mathrm{Rh_R}}$

$\mathrm{d}=\sqrt{2\mathrm{R}}\left[\sqrt{\mathrm{h_T}}+\sqrt{\mathrm{h_R}}\right]$

$\mathrm{d}=\sqrt{2\mathrm{R}}[\sqrt{\mathrm{x}}+\sqrt{160-\mathrm{x}}]$

$\frac{d(d)}{dx}=0$

$\frac{1}{2\sqrt{x}}+\frac{1(-1)}{2\sqrt{160-x}}=0$

$\frac{1}{\sqrt{x}}=\frac{1}{\sqrt{160-x}}$

$x=80\ \mathrm{m}$

$d_{\max}=\sqrt{2\times6400}\left[\sqrt{\frac{80}{1000}}+\sqrt{\frac{20}{1000}}\right]$

$=\frac{80\sqrt{2}\times2\sqrt{80}}{10\sqrt{10}}$

$=8\times2\times\sqrt{2}\times2\sqrt{2}=64\ \mathrm{km}$

Q4 (2)

h : height of antenna

λ : wavelength of signal

$$\mathrm{h}<\lambda$$

$\lambda > \mathrm{h}$

$\lambda > 400 \mathrm{~m}$

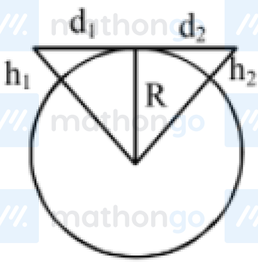
Q5 (224)

$\mathrm{d_m} = \sqrt{2\mathrm{Rh_T}} + \sqrt{2\mathrm{Rh_R}}$

$\mathrm{d_m} = \left(\sqrt{2 \times 6400 \times 10^3 \times 320} + \sqrt{2 \times 6400 \times 10^3 \times 2000}\right) \mathrm{m}$

$\mathrm{d_m} = 224 \mathrm{~km}$

Q6 (4)



$\mathrm{d_1} = \sqrt{2\mathrm{Rh_1}} + \sqrt{2\mathrm{Rh_2}}$

$= \sqrt{2\mathrm{R}}\left(\sqrt{\mathrm{h_1}} + \sqrt{\mathrm{h_2}}\right)$

$= \left(2 \times 6400 \times 10^3\right)^{1/2}(\sqrt{50} + \sqrt{80})$

$= 3578(7.07 + 8.94)$

$= 57.28 \mathrm{Km}$

Q7 (2)

$\mathrm{W_m} = 12560 = 2\pi \mathrm{f_m}$

$\mathrm{f_m} = \frac{12560}{2\pi}$

$= 2000 \mathrm{~Hz}$

Ans. 2.00